**Student Undertaking for Ethical Academic Practice**

**Vidyalankar Institute of Technology, Mumbai**

Assignment No: 10

Branch: CMPN Vertical: PC_PCC
Semester: V Division: A
Subject Name: SE Subject Code: PCCE12T

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SUTKAR

CMPN-A

SE: CDL Assignment 10

Risk Management

Q.1] Prepare an RMM plan for the e-commerce application:-

→ Risk Identification: The e-commerce application may not handle the expected high-user traffic during the festive sale. This could cause slow response times, application crashes, server failures, failed transactions, and loss of custome

→ RMMM stands for Risk Mitigation, Monitoring, and Management.

| | Component | Plan |
|---|---|---|
| 1. | Risk | Server overload or application failure due to high traffic during the festival sale. |
| 2. | Mitigation | Conduct load and stress testing before launch. Optimize database queries and application code. Use load balancing, auto-scaling, caching, and adequate server capacity. |
| 3. | Monitoring | Monitor CPU and memory usage, response time, request rate, error rate, database performance, and server health. Configure alerts when performance access pre-defined thresh |
| 4. | Management (contingency) | Scale up server resources immediat Activate additional server instances, use a waiting queue or rate |

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| | |
|---|---|
| | limiting if required, and temporarily disable non-essential features: maintain backups and communicate service disruptions to customers. |

→ Conclusion:- An effective RMM plan helps identify performance risks early, reduce the likelihood of failure, and minimize the impact of unexpected traffic during the festive sale.

Q2] Calculate Risk Exposure (RE)

→ Formula:-

RE = Probability × potential loss

→ Step1 : Calculate RE for each risk.

① Data migration failure.

Given; Probability = 0.12 , loss = ₹.7,50,000

RE = 0.12 × 7,50,000

**RE = ₹ 90,000**

② Security: vulnerability

Given: Probability = 0.18 , loss = ₹ 6,00,000

**RE = ₹ 1,08,000**

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Page No.:
Date: / /

③ Schedule overrun

Given: Probability = 0.35, Loss = ₹ 2,50,000

RE = 0.35 × 2,50,000

**RE = ₹87,500**

→ Step 2: Ranking the risks in decreasing order of R

| Rank | Risk | Probability | Loss | Risk Exposure |
|---|---|---|---|---|
| 1 | Security vulnerability | 0.18 | ₹6,00,000 | ₹1,08,000 |
| 2 | Data migration failure | 0.12 | ₹7,50,000 | ₹90,000 |
| 3. | Schedule overrun | 0.35 | ₹2,50,000 | ₹87,500 |

→ Step 3: Calculate total Risk Exposure

Total RE = 90,000 + 1,08,000 + 87,500

**RE = ₹2,85,500**

→ Step 4: Identify the highest-priority risk

The security vulnerability should receive the

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highest priority because it has the highest Risk Exposure of ₹1,08,000. The company should prioritize security audits, vulnerability scanning, penetration testing, and timely patching to reduce the risk.